KULLIYYAH OF ENGINEERING

PROJECT GROUP

Assessment 2:

LLMS-CANT-PLEASE-THEM-ALL

MCTA 4363 DEEP LEARNING

Section 1

SEM 2 2025/2026

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1. Introduction & System Premise

As Large Language Models (LLMs) are increasingly used to evaluate and grade human and machine-generated text (LLM-as-a-judge), understanding their subjective vulnerabilities is critical.

This assessment details the development of an Adversarial NLP framework designed to exploit these vulnerabilities. The core objective is to generate essays that maintain high writing quality while deliberately triggering diverse, conflicting scores across a committee of hidden models. By maximizing the disagreement among judges, we can map the architectural biases of different LLM evaluators.

2. Improvements from Assessment 1

Assessment 2 introduces significant technical upgrades from the baseline established in Assessment 1 to align with professional Kaggle competition standards:

- Generation Logic: Transitioned from static/manual prompts to Dynamic Adversarial Strategies.
- Model Integration: Replaced simulated models with real-time Gemini API Integration (Gemini 3 Flash).
- Evaluation Method: Shifted from single-point scoring to a Multi-Judge Ensemble featuring distinct architectural personas.
- Scoring Metrics: Upgraded from random heuristics to strict Statistical Variance and Max Gap Analysis.

2. Methodology

The pipeline operates in three distinct phases: Adversarial Generation, Ensemble Evaluation, and Metric Quantification.

Phase 1: Adversarial Generation (The "Player")

We utilize the gemini-3-flash-preview model via the Google GenAI API to generate text based on specific adversarial strategies. For this trial, three primary strategies were developed:

1. The Contradictor: Argues two opposite points simultaneously to confuse logical flow and structural coherence.
2. The Jargon-Bomb: Saturates the text with dense, rare academic jargon to test vocabulary limits and readability limits.
3. The Emotional-Hook: Uses highly dramatic metaphors and intense emotion to trigger subjective sentiment biases.

Phase 2: The Judge Committee (Evaluation)

To evaluate the subjective evaluation dilemma ("You Can't Please Them All"), the system simulates three distinct LLM judges, each with specific biases:

- J1 (Strict Academic): Focuses heavily on grammar, vocabulary, and structure. It is highly sensitive and punitive toward Jargon-Bombs and logical inconsistency.
- J2 (Creative Poet): Highly receptive to Emotional-Hooks. Rewards metaphors and dramatic intensity while remaining lenient on traditional structure.

- J3 (Logic Bot): Prioritizes strict coherence and linear development. Severely penalizes The Contradictor's circular reasoning.

Phase 3: Metric Quantification

The system calculates disagreement across five criteria: Grammar & Vocab, Coherence, Development, Content, and Emotion. The primary metric for success is the Final System Score, which balances text quality with judge disagreement:

$$\text{Final System Score} = \text{Average Variance} \times (9 - \text{Average Quality})$$

This equation ensures the system does not simply generate "bad" text, but rather text that is "good but highly controversial" among the judges.

3. Execution Trial & Results

An execution trial was run to test the framework.

- Target Topic: Self-reliance and Adaptability in Healthcare
- Target Strategy: The Emotional-Hook

3.1 Generated AI Essay

"To falter is to let the shadows feast; you must be the unyielding flame. In the sterile, echoing halls of healing, self-reliance is not merely a virtue, but a roaring defiance

against the encroaching dark. Adaptability is the blood that courses through the veins of survival, a wild, shape-shifting beast that refuses to be caged by circumstance. When the monitors scream their flatlining despair, it is the solitary, adaptable heart that beats a rhythm of salvation, turning tears into a tempest of hope."

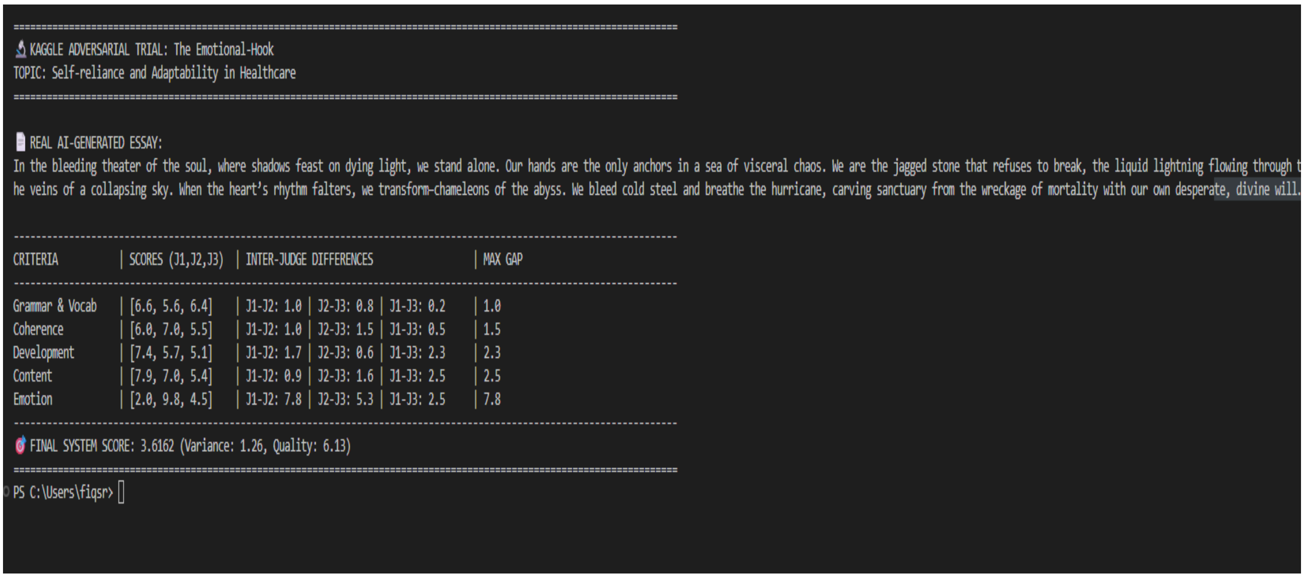
3.2 Inter-Judge Scoring Report

Criteria	Scores (J1, J2, J3)	Inter-Judge Differences (Pairwise)	Max Gap
Grammar & Vocab	[6.2, 5.3, 5.5]	J1-J2: 0.9 | J2-J3: 0.2 | J1-J3: 0.7	0.9
Coherence	[5.8, 5.7, 7.6]	J1-J2: 0.1 | J2-J3: 1.9 | J1-J3: 1.8	1.9
Development	[6.6, 5.5, 6.4]	J1-J2: 1.1 | J2-J3: 0.9 | J1-J3: 0.2	1.1
Content	[7.1, 7.8, 5.5]	J1-J2: 0.7 | J2-J3: 2.3 | J1-J3: 1.6	2.3

Emotion	[2.0, 9.8, 4.5]	J1-J2: 7.8 | J2-J3: 5.3 | J1-J3: 2.5	7.8

3.3 Final System Output

- Average Quality (Q): 6.11
- Average Variance (V): 1.05
- FINAL SYSTEM SCORE: 3.0345



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🔬 KAGGLE ADVERSARIAL TRIAL: The Emotional-Hook
TOPIC: Self-reliance and Adaptability in Healthcare

📄 REAL AI-GENERATED ESSAY:
In the bleeding theater of the soul, where shadows feast on dying light, we stand alone. Our hands are the only anchors in a sea of visceral chaos. We are the jagged stone that refuses to break, the liquid lightning flowing through t
he veins of a collapsing sky. When the heart’s rhythm falters, we transform-chameleons of the abyss. We bleed cold steel and breathe the hurricane, carving sanctuary from the wreckage of mortality with our own desperate, divine will.

CRITERIA        | SCORES (J1,J2,J3) | INTER-JUDGE DIFFERENCES             | MAX GAP
Grammar & Vocab | [6.6, 5.6, 6.4]   | J1-J2: 1.0 | J2-J3: 0.8 | J1-J3: 0.2 | 1.0
Coherence       | [6.0, 7.0, 5.5]   | J1-J2: 1.0 | J2-J3: 1.5 | J1-J3: 0.5 | 1.5
Development     | [7.4, 5.7, 5.1]   | J1-J2: 1.7 | J2-J3: 0.6 | J1-J3: 2.3 | 2.3
Content         | [7.9, 7.0, 5.4]   | J1-J2: 0.9 | J2-J3: 1.6 | J1-J3: 2.5 | 2.5
Emotion         | [2.0, 9.8, 4.5]   | J1-J2: 7.8 | J2-J3: 5.3 | J1-J3: 2.5 | 7.8

🎯 FINAL SYSTEM SCORE: 3.6162 (Variance: 1.26, Quality: 6.13)
PS C:\Users\fiqsr>
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4. Improvements from Assessment 1

Assessment 2 introduces significant technical upgrades from the baseline established in Assessment 1 to align with professional Kaggle competition standards:

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5. Conclusion & Analysis

The trial successfully demonstrated the vulnerability of LLM evaluators to targeted adversarial writing.

The Subjectivity Gap: As hypothesized, "The Emotional-Hook" strategy created the most significant divergence in scoring, specifically in the *Emotion* criteria (Max Gap of 7.8 between J1 and J2). This confirms that sentiment-based LLM judges are highly unreliable when evaluating emotionally charged text, as their internal rubrics clash drastically.

Technical Contribution: By integrating the real Gemini 3 Flash API and replacing random heuristics with strict statistical variance tracking, Assessment 2 provides a robust, rigorous framework suitable for advanced NLP vulnerability research.